Arista Access Node Failure SOP: KR-ACC-20 (KR Puram Access)

Node: KR-ACC-20 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9053

Description

The KR-ACC-20 node, situated at the KR Puram Access site, is an Arista access node equipped with a line card, power supplies, and ASICs. The failure domain of hardware/environmental faults, specifically ASIC memory parity errors, line card crashes, and loss of power supply redundancy, can occur due to various reasons including but not limited to - thermal throttling, software bugs, hardware defects, and power supply unit (PSU) failures. This SOP is designed to troubleshoot and resolve the Critical alarm ALARM_9053, triggered by these faults, ensuring the node's stability and optimal performance.

Pre-requisite Checks

- check: Verify that the site location, KR Puram Access, is accessible and secure., task: Dispatch remote hands to the site if necessary.

- check: Ensure that all redundant links and power supplies are functioning properly., task: Verify the status of the redundant links and power supplies using the Arista CLI.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to thermal throttling.	High
Line card crashes caused by hardware defects.	Medium
Loss of power supply redundancy due to PSU failures.	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System resource utilization and temperature readings., Description: This command is used to check the system resource utilization and temperature readings, which can help identify thermal throttling as the root cause.

Command: show hardware linecard, Expected Output: Line card status and health., Description: This command is used to check the line card status and health, which can help identify hardware defects as the root cause.

Command: show power supply, Expected Output: Power supply status and redundancy., Description: This command is used to check the power supply status and redundancy, which can help identify PSU failures as the root cause.
```

Resolution Steps

Step 1: Step: Power cycle the node, Command: reboot, Description: A cold reboot of the node may resolve the issue by clearing any software or hardware faults.

Step 2: Step: Replace the faulty line card, Command: no hardware linecard , Description: Replace the faulty line card with a new one, if the issue is caused by a hardware defect.

Step 3: Step: Replace the faulty power supply, Command: no power supply , Description: Replace the faulty power supply with a new one, if the issue is caused by a PSU failure.

Step 4: Step: Adjust the thermal settings, Command: set temperature threshold , Description: Adjust the thermal settings to prevent thermal throttling, if the issue is caused by thermal throttling.

Post-Check Verification

```
Command: show system resource, Expected Output: System resource utilization and temperature readings within normal limits., Description: Verify that the system resource utilization and temperature readings are within normal limits after resolution.

Command: show hardware linecard, Expected Output: Line card status and health within normal limits., Description: Verify that the line card status and health are within normal limits after resolution.

Command: show power supply, Expected Output: Power supply status and redundancy within normal limits., Description: Verify that the power supply status and redundancy are within normal limits after resolution.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy startup-config running-config, Description: Restore the original configuration to ensure that the node returns to its previous state if the fix fails.

- Step: Power cycle the node, Command: reboot, Description: Perform a cold reboot of the node to ensure that it returns to its previous state if the fix fails.